ChatGPT

Please prepare 2 analytical, 2 descriptive, 2 short questions from each ppt files. each questions carry 9 marks, duration for answer is 35 minutes for each questions, and each questions may consists of part a, b, c and d.

Certainly. I have prepared **2 Analytical, 2 Descriptive, and 2 Short Questions from each of the 6 uploaded PPT files**. Thus, there are **36 questions in total**.

Question format:

- Each question = **9 marks**
- Time = **35 minutes per question**
- Questions may contain **(a), (b), (c), (d)** parts
- The questions are based specifically on the uploaded PPT materials and their terminology. For example, the Introduction PPT covers classification, regression, clustering, dimensionality reduction, association, and learning methods.
- The Naïve Bayesian PPT emphasizes conditional independence, probability estimation, the tennis example, and performance limitations.
- The MAP/MLE PPT covers model selection, MAP, MLE, coin tossing, and probabilistic linear regression.
- The MLP PPT covers perceptrons, multilayer structures, XOR, learning, and error backpropagation.
- The two SVM PPTs cover maximum-margin linear SVM, support vectors, soft margin, nonlinear SVM, kernel trick, and kernel functions.

1. Introduction to Machine Learning

Source: *0. Introduction.pptx*

A. Analytical Questions

Q1. Classification, Regression and Clustering — 9 Marks

Time: 35 minutes

Consider the following three machine-learning applications:

1. Identifying whether an email is spam or non-spam.
2. Predicting the stock price for the next day.

3. Grouping customers according to their purchasing behavior without predefined labels.

(a) Identify the appropriate machine-learning problem for each application. **(3 marks)**
(b) Explain whether the training data for each problem should be labeled or unlabeled. **(2 marks)**
(c) Explain how the expected output differs among the three problems. **(2 marks)**
(d) Give one reason why selecting the correct type of ML problem is important before designing the learning system. **(2 marks)**

Q2. Selecting an Appropriate Learning Method — 9 Marks

Time: 35 minutes

A university wants to develop four ML systems:

- A system to classify students as pass/fail.
- A system to group students according to learning behavior when no class labels are available.
- A system that learns how to win a game using rewards.
- A system where only some training examples are labeled.

(a) Select the appropriate learning method for each system. **(4 marks)**
(b) Explain the difference between supervised and unsupervised learning in this situation. **2 marks)**
(c) Explain why the fourth system can be considered semi-supervised learning. **(2 marks)**
(d) State the role of reward in reinforcement learning. **(1 mark)**

B. Descriptive Questions

Q3. Types of Machine-Learning Problems — 9 Marks

Time: 35 minutes

(a) Define **classification** and explain how a classifier handles a query. **(3 marks)**
(b) Describe the concept of **regression** and explain the role of $x$and y. **(2 marks)**
(c) Describe **clustering** and explain how it differs from classification. **(2 marks)**
(d) Briefly explain **dimension reduction**. **(2 marks)**

The PPT describes classification as learning from labeled data and regression as finding a functional relationship between input and a real-valued output.

Q4. Parametric, Semiparametric and Nonparametric Learning — 9 Marks

Time: 35 minutes

(a) Describe parametric learning and the role of a global model. **(3 marks)**
(b) Explain semiparametric learning and the use of local models. **(2 marks)**
(c) Describe the basic idea of nonparametric learning. **(2 marks)**
(d) Explain why nonparametric learning is also associated with lazy, memory-based, case-based, or instance-based learning. **(2 marks)**

C. Short Questions

Q5. Basic Concepts — 9 Marks

Time: 35 minutes

(a) Define Machine Learning. **(2 marks)**
(b) Mention two applications of Machine Learning. **(2 marks)**
(c) What is classification? **(2 marks)**
(d) What is reinforcement learning? **(3 marks)**

Q6. Learning Categories — 9 Marks

Time: 35 minutes

(a) What is supervised learning? **(2 marks)**
(b) What is unsupervised learning? **(2 marks)**
(c) What is semi-supervised learning? **(2 marks)**
(d) Mention two differences between supervised and reinforcement learning. **(3 marks)**

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2. Naïve Bayesian Classifier

Source: *5. Naive Bayesian.ppt*

A. Analytical Questions

Q7. Naïve Bayesian Tennis Prediction — 9 Marks

Time: 35 minutes

Using the tennis-playing dataset given in the PPT, consider the query:

Sunny, Mild, High, Strong

(a) Determine the prior probability of the classes **Play = Yes** and **Play = No** from the given 14 training instances. **(2 marks)**
(b) Determine the required conditional probabilities for the four attributes for each class. **(3**

marks)
(c) Using the Naïve Bayesian assumption, calculate the relative probability for both classes. **(3 marks)**
(d) Predict whether the person will play tennis. **(1 mark)**

The PPT explicitly uses this tennis query and asks whether the person will play tennis.

Q8. Conditional Independence and NBC — 9 Marks

Time: 35 minutes

A Naïve Bayesian classifier uses three attributes A_1,A_2,$A_3$to predict classes $C_1$and C_2.

(a) Explain the conditional-independence assumption used by NBC. **(3 marks)**
(b) Explain how the assumption reduces the probability estimation problem. **(2 marks)**
(c) Discuss what may happen if the attributes are strongly dependent on one another. **(2 marks)**
(d) Explain why NBC may still work well even when the independence assumption is violated. **(2 marks)**

The PPT notes that conditional independence is often violated, but NBC can nevertheless work surprisingly well.

B. Descriptive Questions

Q9. Formal Description of Naïve Bayesian Classifier — 9 Marks

Time: 35 minutes

(a) Describe the target function $f:X \to V$used in the formal description of NBC. **(2 marks)**
(b) Explain how NBC determines the most probable class for a query $x = (a_1, a_2, \ldots, a_n)$. **(3 marks)**
(c) Explain the Naïve Bayesian independence assumption. **(2 marks)**
(d) Explain how the probabilities are estimated from given training examples. **(2 marks)**

Q10. Performance and Applications of NBC — 9 Marks

Time: 35 minutes

(a) Describe the main performance characteristics of Naïve Bayesian Classification. **(3 marks)**
(b) Under what training-data conditions is NBC recommended according to the PPT? **(2 marks)**
(c) Mention and explain two successful applications of NBC. **(2 marks)**
(d) Explain why NBC can give poor performance when the number of relevant training instances is small. **(2 marks)**

The PPT specifically identifies diagnosis and spam/non-spam document classification as successful applications and discusses poor performance with small data.

C. Short Questions

Q11. NBC Fundamentals — 9 Marks

Time: 35 minutes

(a) What is a Naïve Bayesian Classifier? **(2 marks)**
(b) What is meant by conditional independence? **(2 marks)**
(c) What is a prior probability? **(2 marks)**
(d) Why is NBC called "Naïve"? **(3 marks)**

Q12. Probability Estimation — 9 Marks

Time: 35 minutes

(a) What is a class prior? **(2 marks)**
(b) What is a conditional probability? **(2 marks)**
(c) What problem occurs when the available training data for a particular value is very small? **(2 marks)**
(d) Explain the roles of $p$and $m$in the estimation method presented in the PPT. **(3 marks)**

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3. Model Selection: MAP and MLE

Source: *6. MAP_MLE.ppt*

A. Analytical Questions

Q13. MAP Estimation Using Coin Tossing — 9 Marks

Time: 35 minutes

Five coins are available in a pocket:

- One coin c_1: two heads,
- Three coins c_2: one head and one tail,
- One coin c_3: two tails.

One coin is selected randomly and tossed once. The observed result is **Head**.

(a) Identify the sample/data and the candidate models. **(2 marks)**
(b) Determine the prior probability of selecting each candidate coin. **(2 marks)**

(c) Calculate the posterior probability conceptually using Bayes' rule. **(3 marks)**
(d) Identify the MAP estimator and justify your answer. **(2 marks)**

The PPT presents this exact coin-tossing example and identifies the MAP estimator as c_3.

Q14. MLE and Linear Regression — 9 Marks

Time: 35 minutes

A set of sample points is assumed to be generated from a linear model with random noise following a normal distribution.

(a) Explain the probabilistic assumption used for the data generation. **(2 marks)**
(b) Explain how MLE can be used to select the linear model. **(2 marks)**
(c) Explain the relationship between maximizing likelihood and minimizing squared error. **(3 marks)**
(d) State the final relationship between the MLE solution and the squared-error linear regression solution. **(2 marks)**

The PPT states that the data are generated from a linear model with normally distributed random noise and that the MLE and squared-error solutions are equivalent for linear regression.

B. Descriptive Questions

Q15. MAP versus MLE — 9 Marks

Time: 35 minutes

(a) Define model selection. **(2 marks)**
(b) Describe the Maximum A Posteriori (MAP) approach. **(3 marks)**
(c) Describe the Maximum Likelihood Estimator (MLE) approach. **(2 marks)**
(d) Explain the major conceptual difference between MAP and MLE. **(2 marks)**

The PPT defines MAP as choosing the model with maximum posterior probability and MLE as choosing the model with maximum likelihood.

Q16. Two Approaches to Model Selection — 9 Marks

Time: 35 minutes

(a) Explain model selection based on an error function. **(2 marks)**
(b) Explain model selection based on probability. **(3 marks)**
(c) Mention the optimization methods associated with these approaches in the PPT. **(2 marks)**

(d) Explain why probability-based model selection can be useful when choosing a model for sample data. **(2 marks)**

C. Short Questions

Q17. Basic MAP and MLE Concepts — 9 Marks

Time: 35 minutes

(a) What is MAP? **(2 marks)**
(b) What is MLE? **(2 marks)**
(c) What is likelihood? **(2 marks)**
(d) Why does MLE not explicitly use the prior probability of a model? **(3 marks)**

Q18. Model Selection — 9 Marks

Time: 35 minutes

(a) What is a sample? **(2 marks)**
(b) What is a candidate model? **(2 marks)**
(c) What is the purpose of model selection? **(2 marks)**
(d) State the two general objectives: minimizing error and maximizing probability. **(3 marks)**

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4. Neural Network: Multi-Layer Perceptron

Source: *9. Multi Layer Perceptron.ppt*

A. Analytical Questions

Q19. Perceptron for AND and OR — 9 Marks

Time: 35 minutes

Consider a perceptron with two binary inputs x_1,x_2.

(a) Using $w_1 = 1.0$,$w_2 = 1.0$,$w_0 = -1.5$, calculate $S$and output $y$for all four input combinations and verify the AND operation. **(4 marks)**
(b) Using $w_1 = 1.0$,$w_2 = 1.0$,$w_0 = -0.5$, calculate the outputs for all four combinations and verify the OR operation. **(3 marks)**
(c) Explain how changing only the bias/threshold changes the logical operation. **(2 marks)**

The PPT gives these exact AND and OR weights and truth-table calculations.

Q20. MLP Solution for XOR — 9 Marks

Time: 35 minutes

A single perceptron cannot represent the XOR decision boundary.

(a) Explain why XOR creates a problem for a single perceptron. **(2 marks)**
(b) Describe how a multilayer perceptron can solve XOR. **(2 marks)**
(c) Explain the roles of the hidden and output layers in solving XOR. **(3 marks)**
(d) Explain why multiple perceptrons can produce a nonlinear decision boundary. **(2 marks)**

The PPT explicitly uses XOR as an example of a problem that can be solved by a neural network and provides an MLP structure for XOR.

B. Descriptive Questions

Q21. Structure of a Multilayer Perceptron — 9 Marks

Time: 35 minutes

(a) Describe the input layer of an MLP. **(2 marks)**
(b) Describe the hidden layer and explain why several hidden layers may be used. **(3 marks)**
(c) Describe the output layer and its relationship to the number of outputs. **(2 marks)**
(d) Mention two other names used for multilayer perceptron in the PPT. **(2 marks)**

The PPT states that the input layer passes input values forward, hidden layers may be multiple, and the output-layer node count corresponds to the number of outputs.

Q22. Error Backpropagation Learning — 9 Marks

Time: 35 minutes

(a) Explain the basic idea of error backpropagation. **(2 marks)**
(b) Explain why an error function is required in neural-network learning. **(2 marks)**
(c) Explain how the chain rule is used in calculating weight changes. **(3 marks)**
(d) Describe the major steps of the backpropagation learning algorithm. **(2 marks)**

The PPT describes backpropagation as minimizing error using input-target pairs and distinguishes weights between output-hidden and hidden-input layers.

C. Short Questions

Q23. Perceptron Fundamentals — 9 Marks

Time: 35 minutes

(a) What is a perceptron? **(2 marks)**
(b) What is the weighted summation function? **(2 marks)**

(c) What is the role of the nonlinear function? **(2 marks)**
(d) Write the general expression for $s$used in the perceptron. **(3 marks)**

The PPT defines the two main perceptron functions as weighted summation and a nonlinear function.

Q24. MLP Capabilities — 9 Marks

Time: 35 minutes

(a) What is an Artificial Neural Network? **(2 marks)**
(b) What is a feed-forward neural network? **(2 marks)**
(c) What type of function can a two-layer perceptron represent according to the PPT? **2 marks)**
(d) Why is a multilayer perceptron useful for nonlinear problems? **(3 marks)**

The PPT states that Boolean functions can be represented by a two-layer perceptron, continuous functions by a two-layer perceptron, and arbitrary functions by a three-layer perceptron.

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5. Support Vector Machine-I: Linear SVM

Source: *12-1. Support Vector Machine-1.ppt*

A. Analytical Questions

Q25. Maximum-Margin Boundary — 9 Marks

Time: 35 minutes

For a two-class classification problem, a linear decision boundary is represented by:

$$w^T x + b = 0$$

(a) Explain what the positive and negative regions of the boundary represent. **(2 marks)**
(b) Define the two margin boundaries using the formulation presented in the PPT. **(2 marks)**
(c) Derive the expression for the margin width from the distance between the two margin boundaries. **(3 marks)**
(d) Explain why maximizing the margin is used as the objective of linear SVM. **(2 marks)**

The PPT defines the decision boundary and the margin boundaries as $wx + b = 1$and $wx + b = -1$, and derives the margin from their distance.

Q26. SVM Classification Constraints — 9 Marks

Time: 35 minutes

Suppose the class labels are $y_i \in \{+, 1 - 1\}$.

(a) Explain the purpose of assigning $+1$and -1class labels. **(2 marks)**
(b) Write the SVM constraint for the training examples. **(2 marks)**
(c) Explain how the constraint applies to positive and negative examples. **(3 marks)**
(d) Explain how these constraints help obtain the maximum-margin classifier. **(2 marks)**

The PPT gives the condition $y_i(w \cdot x_i + b) \geq 1$for all training examples.

B. Descriptive Questions

Q27. Support Vectors and Margin — 9 Marks

Time: 35 minutes

(a) Define the margin of a linear classifier. **(2 marks)**
(b) Define support vectors. **(3 marks)**
(c) Explain why support vectors are particularly important in SVM learning. **(2 marks)**
(d) Explain what happens to the influence of non-support-vector training examples. **(2 marks)**

The PPT states that support vectors are the data points against which the margin pushes and notes that only support vectors are important in the resulting classifier.

Q28. Linear SVM with Soft Margin — 9 Marks

Time: 35 minutes

(a) Explain why a soft-margin SVM is required for a non-linearly separable/noisy dataset. **(2 marks)**
(b) Describe the role of the slack variable. **(2 marks)**
(c) Explain the role of parameter C. **(3 marks)**
(d) Describe the consequences of very large and very small C. **(2 marks)**

The PPT explains that soft-margin SVM allows error and that Ccontrols the trade-off between margin and error.

C. Short Questions

Q29. Linear SVM Fundamentals — 9 Marks

Time: 35 minutes

(a) What is a linear SVM? **(2 marks)**
(b) What is a decision boundary? **(2 marks)**
(c) What is a margin? **(2 marks)**
(d) What is a support vector? **(3 marks)**

Q30. SVM Optimization — 9 Marks

Time: 35 minutes

(a) What is the objective of linear SVM learning? **(2 marks)**
(b) What is the role of $w$and b? **(2 marks)**
(c) What is quadratic programming in the context of SVM? **(2 marks)**
(d) Why is the dual formulation useful for learning SVM? **(3 marks)**

The PPT discusses quadratic programming and the dual formulation as approaches for learning the maximum-margin boundary.

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6. Support Vector Machine-II: Nonlinear SVM and Kernel Trick

Source: *12-2. Support Vector Machine-2.ppt*

A. Analytical Questions

Q31. Nonlinear SVM Using Feature Mapping — 9 Marks

Time: 35 minutes

A dataset cannot be separated by a linear boundary in its original input space.

(a) Explain why a nonlinear decision boundary is required. **(2 marks)**
(b) Explain the idea of mapping the original data into a higher-dimensional feature space. **(2 marks)**
(c) Explain how a linear boundary in the higher-dimensional space can correspond to a nonlinear boundary in the original space. **(3 marks)**
(d) State the main computational difficulty associated with explicitly constructing a high-dimensional mapping. **(2 marks)**

The PPT presents exactly this strategy: map data to a higher-dimensional space and then find a linear boundary there.

Q32. Kernel Trick — 9 Marks

Time: 35 minutes

Consider a nonlinear SVM using a kernel function $K(x_i, x_j)$.

(a) Explain why inner products of mapped data are important in nonlinear SVM. **(2 marks)**
(b) Define the kernel function according to the PPT. **(2 marks)**

(c) Explain how the Kernel Trick avoids explicitly calculating the transformation $\Phi(x)$. **(3 marks)**
(d) Explain why this can save computational time. **(2 marks)**

The PPT explains that a kernel corresponds to the dot product of feature vectors in an expanded feature space and that a well-designed transformation allows the inner product to be evaluated efficiently.

B. Descriptive Questions

Q33. Common Kernel Functions — 9 Marks

Time: 35 minutes

(a) Explain the purpose of a kernel function in nonlinear SVM. **(2 marks)**
(b) Describe the Linear Kernel. **(2 marks)**
(c) Describe Polynomial and Homogeneous Polynomial Kernels. **2 marks)**
(d) Describe the Gaussian/RBF and Sigmoid kernels. **(3 marks)**

The PPT lists linear, homogeneous polynomial, polynomial, Gaussian/RBF, and sigmoid kernels as commonly used kernel functions.

Q34. Mercer's Condition and SVM Procedure — 9 Marks

Time: 35 minutes

(a) Explain the purpose of Mercer's condition in relation to kernel functions. **(3 marks)**
(b) Explain the relationship $K(x, y) = \Phi(x) \cdot \Phi(y)$. **(2 marks)**
(c) Describe the main steps for using an SVM. **(2 marks)**
(d) Explain the role of k-fold validation in selecting kernel parameters and C. **(2 marks)**

The PPT presents Mercer's condition and the practical SVM procedure, including kernel selection, selection of C, k-fold validation, quadratic programming, and construction of the discriminant function.

C. Short Questions

Q35. Nonlinear SVM Fundamentals — 9 Marks

Time: 35 minutes

(a) What is nonlinear SVM? **(2 marks)**
(b) Why is higher-dimensional feature space used? **(2 marks)**
(c) What is a kernel? **(2 marks)**
(d) What is the Kernel Trick? **(3 marks)**

Q36. Kernels and SVM — 9 Marks

Time: 35 minutes

(a) Mention four commonly used kernel functions. **(2 marks)**
(b) What is an RBF kernel? **(2 marks)**
(c) State one advantage and one disadvantage of SVM. **(2 marks)**
(d) Mention the major steps involved in using an SVM. **(3 marks)**

The PPT lists several advantages and disadvantages and gives the practical steps for applying SVM, including kernel selection, Cselection, quadratic programming, and construction of the discriminant function.

Summary of the Question Distribution

PPT	Analytical	Descriptive	Short	Total
0. Introduction	2	2	2	**6**
5. Naïve Bayesian	2	2	2	**6**
6. MAP & MLE	2	2	2	**6**
9. Multi-Layer Perceptron	2	2	2	**6**
12-1. SVM-I	2	2	2	**6**
12-2. SVM-II	2	2	2	**6**
Total	**12**	**12**	**12**	**36**

Every question carries 9 marks and is allocated 35 minutes. The analytical questions emphasize calculation, interpretation, derivation, application, or reasoning; descriptive questions emphasize explanation and discussion; and short questions focus on concise conceptual understanding.

DeepSeek

Please prepare 2 analytical, 2 descriptive, 2 short questions from each ppt files. each questions carry 9 marks, duration for answer is 35 minutes for each questions, and each questions may consists of part a, b, c and d.

Machine Learning - Comprehensive Question Bank

Based on: 0. Introduction.pptx

Analytical Questions

Question 1: Types of ML Problems Analysis (9 marks)

(a) Distinguish between supervised, unsupervised, and reinforcement learning by providing two real-world applications for each. (3 marks)

(b) For the following scenario, identify the ML problem type and justify your answer: "A bank wants to automatically approve or reject loan applications based on applicant data and historical loan repayment records." (2 marks)

(c) Explain why classification and regression are both considered supervised learning problems despite producing different types of outputs. Provide one example of each where the output differs. (2 marks)

(d) If you were tasked with building a system that groups customer purchase patterns without any pre-existing labels, which type of ML problem would you solve and why? (2 marks)

Question 2: Parametric vs Nonparametric Methods (9 marks)

(a) Define parametric, semiparametric, and nonparametric learning methods with clear distinctions. (3 marks)

(b) Compare the memory requirements and prediction time complexity of parametric and nonparametric methods. (2 marks)

(c) For a k-Nearest Neighbor classifier, explain why it is called a "lazy learner" and discuss one advantage and one disadvantage of this approach. (2 marks)

(d) Consider a dataset with 10,000 samples and 100 features. Would you prefer a parametric or nonparametric method? Justify with computational and accuracy considerations. (2 marks)

Descriptive Questions

Question 3: Machine Learning Fundamentals (9 marks)

(a) Define machine learning and explain the core objective of ML programs in your own words. (2 marks)

(b) List and describe four key application areas of machine learning with specific examples for each. (4 marks)

(c) Explain why machine learning overlaps with statistics but differs in its primary focus. (2 marks)

(d) Why can ML incorporate more heuristic knowledge than traditional statistical methods? Provide an example. (1 mark)

Question 4: History and Evolution of Machine Learning (9 marks)

(a) Create a timeline showing the major developments in machine learning from the 1950s to the 2000s. (4 marks)

(b) Explain the significance of Minsky and Papert's work on the Perceptron. What impact did it have on the field? (2 marks)

(c) What was the "knowledge acquisition bottleneck" in expert systems and how did machine learning help address it? (2 marks)

(d) Briefly explain the resurgence of neural networks in the 1980s and what enabled it. (1 mark)

Short Questions

Question 5: ML Basic Concepts (9 marks)

(a) What are the three main types of ML problems based on the nature of the learning signal? (2 marks)

(b) Define "meta-learning" and provide one example method. (2 marks)

(c) List the mathematical prerequisites for this machine learning course. (3 marks)

(d) What is the evaluation breakdown for this course (HW, Mid, Final)? (2 marks)

Question 6: Application Identification (9 marks)

Identify the ML problem type and learning method for each application:

(a) A spam filter that learns from thousands of labeled emails. (1.5 marks)

(b) A movie recommendation system that suggests films based on viewing patterns without explicit ratings. (1.5 marks)

(c) A self-driving car that learns to navigate through trial and error with reward signals. (1.5 marks)

(d) A system that predicts stock prices based on historical market data. (1.5 marks)

(e) A facial recognition system that identifies individuals in photos. (1.5 marks)

(f) A system that finds groups of similar documents in a large text corpus. (1.5 marks)

Based on: 9. Multi Layer Perceptron.ppt

Analytical Questions

Question 7: Perceptron Capabilities and Limitations (9 marks)

(a) What is the mathematical function computed by a single perceptron? Explain the two-step evaluation process. (2 marks)

(b) Show using a truth table that a single perceptron can implement the AND and OR logical functions. (3 marks)

(c) Explain why a single perceptron cannot solve the XOR problem. What is the fundamental limitation? (2 marks)

(d) Prove mathematically that the XOR function is not linearly separable. (2 marks)

Question 8: Error Back Propagation Analysis (9 marks)

(a) Derive the weight update rule for output layer weights in a multilayer perceptron using gradient descent. Start from the error function $E = ½ \Sigma(t_k - o_k)^2$. (4 marks)

(b) Explain the chain rule application for computing $\partial E/\partial w_{ji}$ for weights between hidden and input layers. (3 marks)

(c) Why is backpropagation considered a "generalization" of the delta rule? (1 mark)

(d) What is the vanishing gradient problem and when does it typically occur? (1 mark)

Descriptive Questions

Question 9: Multilayer Perceptron Capabilities (9 marks)

(a) State the universality theorem for multilayer perceptrons. What can a 2-layer and 3-layer perceptron compute? (3 marks)

(b) Draw the architecture of a multilayer perceptron that solves the XOR problem. Label all weights and show the output for each input combination. (4 marks)

(c) Why are hidden layers necessary for solving non-linearly separable problems? (2 marks)

Question 10: Neural Network Design Issues (9 marks)

(a) Discuss the trade-off between model complexity and generalization in neural networks. Use the concepts of underfitting and overfitting. (3 marks)

(b) Explain why early stopping is used during neural network training. How does it help prevent overfitting? (2 marks)

(c) For multi-class classification, how many output nodes are required and what is the recommended target encoding? (2 marks)

(d) Why should regression outputs be normalized into [0,1] when using sigmoid activation in the output layer? (2 marks)

Short Questions

Question 11: MLP Architecture and Components (9 marks)

(a) What does the term "multilayer perceptron" refer to in the context of neural networks? (2 marks)

(b) How many layers does a network with one hidden layer have? Name each layer. (2 marks)

(c) What is the role of the activation function in a perceptron? (2 marks)

(d) If you have a network with 10 inputs, one hidden layer with 5 nodes, and 3 outputs, how many weights need to be learned? Show your calculation. (3 marks)

Question 12: Backpropagation Concepts (9 marks)

(a) Define the error function typically used in backpropagation for a network with multiple outputs. (2 marks)

(b) Why is the gradient of the error with respect to weights computed using the chain rule? (2 marks)

(c) In backpropagation, what distinguishes the weight update rule for output layer weights from hidden layer weights? (3 marks)

(d) What is the role of the learning rate (η) in the backpropagation algorithm? (2 marks)

Based on: 12-1. Support Vector Machine-1.ppt

Analytical Questions

Question 13: Margin Maximization Derivation (9 marks)

(a) Define the margin of a linear classifier. How is it mathematically computed? (2 marks)

(b) Show that the distance between the plus and minus boundaries ($w \cdot x + b = 1$ and $w \cdot x + b = -1$) is $2/||w||$. (4 marks)

(c) Write the optimization problem for finding the maximum margin separating hyperplane. (2 marks)

(d) What are support vectors and why are they important in SVM? (1 mark)

Question 14: SVM Dual Formulation (9 marks)

(a) Write the primal optimization problem for Linear SVM. (2 marks)

(b) Derive the Lagrangian dual formulation from the primal problem. Show the Lagrangian function and the conditions for optimality. (4 marks)

(c) From the dual solution, how do you obtain the weight vector w and bias b? (2 marks)

(d) What is the advantage of solving the dual problem instead of the primal? (1 mark)

Descriptive Questions

Question 15: Soft Margin SVM (9 marks)

(a) Explain the purpose of the soft margin SVM. Why is it necessary for real-world data? (3 marks)

(b) Write the formulation for soft margin SVM including slack variables. Explain the role of the parameter C. (3 marks)

(c) Compare hard margin and soft margin SVM by discussing: (i) when each is appropriate, (ii) their sensitivity to outliers, and (iii) the meaning of C values (large vs small). (3 marks)

Question 16: SVM Decision Boundaries (9 marks)

(a) Given the following linearly separable dataset with two classes, determine the SVM decision boundary. Show all support vectors. (4 marks)

Points: Class A: (1,1), (2,2), (2,0) | Class B: (4,4), (5,5), (5,3)

(b) Explain how the choice of C affects the decision boundary in the presence of noise. (3 marks)

(c) Why does increasing C tend to lead to overfitting in SVM? (2 marks)

Short Questions

Question 17: SVM Fundamentals (9 marks)

(a) What is the objective of Linear SVM? (2 marks)

(b) Define "margin" in the context of SVM. (2 marks)

(c) What are support vectors and how do they determine the decision boundary? (2 marks)

(d) For a dataset with 1000 samples, approximately how many support vectors typically exist in a well-trained SVM? Why? (3 marks)

Question 18: SVM Optimization (9 marks)

(a) Write the constraint that each training example must satisfy in hard margin SVM. (2 marks)

(b) What type of optimization problem does SVM solve? (2 marks)

(c) Why can't the primal formulation be solved directly for large datasets? (2 marks)

(d) What does the condition $\Sigma \alpha_i y_i = 0$ represent in the dual formulation? (3 marks)

Based on: 12-2. Support Vector Machine-2.ppt

Analytical Questions

Question 19: Kernel Trick and Non-linear SVM (9 marks)

(a) Explain the concept of the kernel trick. Why is it considered the "trick" in SVM? (3 marks)

(b) If $x = (x_1, x_2)$ in 2D, derive the explicit mapping $\Phi(x)$ for the kernel $K(x,z) = (x \cdot z)^2$. Show the dimensionality of the mapped space. (3 marks)

(c) Show that using $K(x,z) = (x \cdot z + 1)^2$ is equivalent to mapping to a higher dimensional feature space. (2 marks)

(d) What is the computational advantage of using the kernel trick? (1 mark)

Question 20: Mercer's Condition (9 marks)

(a) State Mercer's condition. What does it guarantee about a kernel function? (3 marks)

(b) Determine whether $K(x,z) = (x \cdot z)^3$ is a valid kernel function. Justify using Mercer's condition. (3 marks)

(c) Given the following kernel matrices, which are valid kernels and why? (3 marks)

$K_1 = [[4, 2], [2, 1]]$
$K_2 = [[1, -1], [-1, 4]]$

Descriptive Questions

Question 21: Kernel Functions and Their Properties (9 marks)

(a) List five commonly used kernel functions and write their mathematical expressions. (4 marks)

(b) Compare the RBF kernel and the Polynomial kernel in terms of: (i) number of parameters, (ii) flexibility, (iii) computational cost. (3 marks)

(c) Explain why choosing an appropriate kernel is the most important design decision in SVM. (2 marks)

Question 22: SVM vs RBF Neural Networks (9 marks)

(a) Draw a diagram showing how an SVM with RBF kernel relates to an RBF neural network. (3 marks)

(b) Discuss the key differences between RBF neural networks and SVM with RBF kernel. (3 marks)

(c) What are the advantages and disadvantages of using SVM compared to neural networks? (3 marks)

Short Questions

Question 23: Kernel Functions (9 marks)

(a) Define a kernel function. (2 marks)

(b) Write the formula for the Gaussian/RBF kernel. (2 marks)

(c) What is the effect of increasing the kernel parameter σ in the RBF kernel? (2 marks)

(d) Why is the RBF kernel the most commonly used kernel in practice? (3 marks)

Question 24: SVM Practical Issues (9 marks)

(a) What are the steps involved in using SVM for a classification problem? (3 marks)

(b) What are two limitations of SVM? (2 marks)

(c) How can you choose the proper parameters (C and kernel parameters) for SVM? (2 marks)

(d) Why might SVM have numerical stability problems? (2 marks)

Based on: 14-1. Deep Learning-Limits of NN.pptx

Analytical Questions

Question 25: Deep vs Shallow Networks (9 marks)

(a) According to the universal approximation theorem, a shallow network can approximate any function. Why then do we need deep networks? (3 marks)

(b) Explain how the required number of nodes can grow exponentially in a shallow network compared to a deep network. Provide an illustrative example. (3 marks)

(c) What are the trade-offs between shallow and deep networks in terms of: (i) number of parameters, (ii) training complexity, (iii) expressiveness? (3 marks)

Question 26: Vanishing Gradient Problem (9 marks)

(a) Define the vanishing gradient problem. Why does it occur in deep neural networks? (3 marks)

(b) Derive the gradient expression for weights in the first hidden layer of a 5-layer network using the chain rule. Explain why this gradient tends to vanish with more layers. (4 marks)

(c) What are two practical consequences of vanishing gradients during training? (2 marks)

Descriptive Questions

Question 27: Deep Learning Challenges (9 marks)

(a) List and explain four major challenges in training deep neural networks. (4 marks)

(b) For each challenge, describe one solution approach. (4 marks)

(c) Which challenge does "internal covariate shift" refer to and what causes it? (1 mark)

Question 28: Deep Network Structures (9 marks)

(a) Distinguish between fully connected and partially connected deep networks. What are the advantages of each? (3 marks)

(b) For what types of data are Convolutional Neural Networks specifically designed? Justify your answer. (3 marks)

(c) Explain the difference between Recurrent Neural Networks and Recursive Neural Networks. (3 marks)

Short Questions

Question 29: Deep Learning Terminology (9 marks)

(a) Define "internal covariate shift." (2 marks)

(b) What is the vanishing gradient problem? (2 marks)

(c) How does dropout help prevent overfitting? (2 marks)

(d) What is batch normalization and why is it used? (3 marks)

Question 30: Deep Network Categorization (9 marks)

Classify each as fully connected or partially connected and explain briefly:

(a) Convolutional Neural Network (2 marks)

(b) Recurrent Neural Network (2 marks)

(c) Deep Belief Network (2 marks)

(d) Stacked Autoencoder (3 marks)

Based on: 14-2. Deep Learning-CNN.pptx

Analytical Questions

Question 31: Convolution Operation (9 marks)

(a) Define the convolution operation in the context of CNNs. Write the mathematical expression for a 2D convolution. (2 marks)

(b) Perform the following convolution:

Input Matrix:

text

1 0 1 1

0 1 0 1

1 0 1 0

0 1 0 1

Kernel:

text

1 0

0 1

Show the output feature map. (4 marks)

(c) How does the size of the output feature map depend on input size, kernel size, stride, and padding? (3 marks)

Question 32: CNN Architectural Components (9 marks)

(a) Compare convolution layers with fully connected layers in terms of: (i) number of parameters, (ii) computation complexity, (iii) generalization capability. (3 marks)

(b) Explain the role of pooling layers in CNN. Why is max pooling preferred over average pooling in many applications? (3 marks)

(c) What is the advantage of using shared weights in convolution layers? (3 marks)

Descriptive Questions

Question 33: CNN Feature Learning (9 marks)

(a) Explain the feature hierarchy learned by CNNs. What types of features are detected in early layers vs later layers? (3 marks)

(b) Using the concept of convolution and pooling, explain why CNNs are translation invariant. (3 marks)

(c) How do CNNs learn features automatically compared to traditional computer vision approaches that use hand-crafted features? (3 marks)

Question 34: CNN Architecture and Training (9 marks)

(a) Draw and label the architecture of a typical CNN including convolution, pooling, and fully connected layers. (3 marks)

(b) In a CNN, how are the kernel values (filter weights) learned? (2 marks)

(c) Why is the number of parameters in a CNN much smaller than in a fully connected network with the same number of neurons? (2 marks)

(d) What is the advantage of this reduced parameter count? (2 marks)

Short Questions

Question 35: CNN Terminology (9 marks)

(a) What is a "kernel" or "filter" in a CNN? (2 marks)

(b) What does the pooling operation do? Provide an example. (2 marks)

(c) Why is CNN considered to have "shared weights"? (2 marks)

(d) What is the purpose of having multiple convolutional layers in a CNN? (3 marks)

Question 36: CNN Applications and Properties (9 marks)

(a) For what type of data are CNNs primarily designed? (1.5 marks)

(b) Give three examples of CNN applications. (3 marks)

(c) Explain one advantage and one disadvantage of CNN compared to fully connected networks. (3 marks)

(d) How does CNN reduce model complexity while maintaining learning capability? (1.5 marks)

Based on: 14-3. Deep Learning-Good Init.pptx

Analytical Questions

Question 37: Autoencoder Dimensionality Reduction (9 marks)

(a) What is an autoencoder and how does it work? Draw its architecture. (3 marks)

(b) Explain the objective function used to train an autoencoder. What is it trying to minimize? (3 marks)

(c) How does an autoencoder perform dimensionality reduction and what is the purpose of the bottleneck layer? (3 marks)

Question 38: Greedy Layer-Wise Pre-training (9 marks)

(a) Explain the concept of greedy layer-wise pre-training. Why is it called "greedy"? (3 marks)

(b) Show step-by-step how a stacked autoencoder is built using greedy layer-wise training. (4 marks)

(c) What is the advantage of using this approach over training all layers simultaneously? (2 marks)

Descriptive Questions

Question 39: Feature Learning with Autoencoders (9 marks)

(a) Compare feature extraction using an autoencoder (stacked) with PCA for dimensionality reduction. Which one can capture non-linear relationships and why? (3 marks)

(b) In the context of facial expression recognition, explain why reducing 65,536 dimensions to 60 dimensions might be beneficial. (3 marks)

(c) How does unsupervised pre-training help with the vanishing gradient problem in deep networks? (3 marks)

Question 40: Pre-training and Fine-tuning Pipeline (9 marks)

(a) Describe the two-phase training process for deep networks: unsupervised pre-training and supervised fine-tuning. (4 marks)

(b) What is the purpose of the pre-training phase? What problem does it address? (3 marks)

(c) Why is fine-tuning necessary after pre-training? (2 marks)

Short Questions

Question 41: Autoencoder Concepts (9 marks)

(a) Define an autoencoder. (2 marks)

(b) What is a "bottleneck" in an autoencoder and why is it important? (2 marks)

(c) What is the difference between a stacked autoencoder and a regular autoencoder? (2 marks)

(d) Name two types of autoencoders other than the basic version. (3 marks)

Question 42: Pre-training Strategies (9 marks)

(a) What is "unsupervised pre-training"? (2 marks)

(b) Why is pre-training beneficial for deep networks? (2 marks)

(c) What is "supervised fine-tuning" and when is it performed? (2 marks)

(d) Name three types of networks that use unsupervised pre-training. (3 marks)

Based on: 14-4. Deep Learning-____.pptx

Analytical Questions

Question 43: Activation Functions and ReLU (9 marks)

(a) Compare the sigmoid function and ReLU as activation functions. Include their mathematical expressions, advantages, and disadvantages. (4 marks)

(b) Explain why ReLU helps mitigate the vanishing gradient problem. (3 marks)

(c) What is the "knockout problem" in ReLU? How can it be addressed? (2 marks)

Question 44: Gradient Descent Variants (9 marks)

(a) Distinguish between batch gradient descent, stochastic gradient descent, and mini-batch gradient descent. For each, state: (i) how the gradient is computed, (ii) convergence properties, (iii) computational efficiency. (5 marks)

(b) What is a typical mini-batch size and how is it chosen? (2 marks)

(c) Why is mini-batch gradient descent often preferred over batch and stochastic methods? (2 marks)

Descriptive Questions

Question 45: Regularization Techniques (9 marks)

(a) Define regularization in the context of deep learning. Why is it needed? (2 marks)

(b) Compare L1 and L2 regularization (weight decay). How does each affect the weights and what are their practical implications? (3 marks)

(c) Explain dropout regularization: how it works during training and testing, and why it prevents overfitting. (3 marks)

(d) How can early stopping be used as a regularization technique? (1 mark)

Question 46: Batch Normalization (9 marks)

(a) What is batch normalization? Explain how it addresses the internal covariate shift problem. (3 marks)

(b) Describe the steps of batch normalization for a training batch. (3 marks)

(c) What are the advantages of batch normalization in terms of: (i) learning speed, (ii) accuracy, (iii) flexibility with data distribution? (3 marks)

Short Questions

Question 47: Deep Learning Optimization (9 marks)

(a) What is the vanishing gradient problem and which activation function helps address it? (2 marks)

(b) Define internal covariate shift. (2 marks)

(c) What is the advantage of using mini-batch gradient descent over full-batch gradient descent? (2 marks)

(d) Why is dropout considered a regularization technique? (3 marks)

Question 48: Solutions to Deep Learning Challenges (9 marks)

Match each challenge with the best solution approach:

Challenge	Solution Options
(a) Vanishing gradient	(i) Dropout, (ii) ReLU, (iii) Batch Normalization, (iv) Pre-training
(b) Internal covariate shift	(i) Dropout, (ii) ReLU, (iii) Batch Normalization, (iv) Pre-training
(c) Overfitting	(i) Dropout, (ii) ReLU, (iii) Batch Normalization, (iv) Pre-training
(d) Hard to optimize	(i) Dropout, (ii) ReLU, (iii) Batch Normalization, (iv) Pre-training

For each answer, provide a brief justification. (9 marks, 2.25 marks each)

Based on: 15. Decision tree _ ID3.ppt

Analytical Questions

Question 49: Entropy Calculation (9 marks)

(a) Define entropy as a measure of disorder. Write the formula for entropy for a set with c classes. (2 marks)

(b) Calculate the entropy for a set with 9 positive examples and 5 negative examples. Show all steps. (3 marks)

(c) After splitting on attribute A, you get: Subset 1: 6 positive, 1 negative; Subset 2: 3 positive, 4 negative. Calculate the entropy after the split. (2 marks)

(d) What is the information gain from this split? (2 marks)

Question 50: ID3 Algorithm Analysis (9 marks)

(a) Write the ID3 algorithm for constructing a decision tree. (4 marks)

(b) For the following dataset, determine which attribute should be selected as the root node. Show calculations for information gain for each attribute. (5 marks)

Example	A	B	C	Class
1	T	T	F	Yes
2	T	F	T	Yes
3	F	T	T	No
4	F	F	F	No

Descriptive Questions

Question 51: Decision Tree Construction (Tennis Example) (9 marks)

(a) Build a decision tree using the ID3 algorithm for the tennis dataset (provided in the slides). Show each step of attribute selection. (6 marks)

(b) What is the predicted outcome for the query: Outlook=Sunny, Temp=Mild, Humidity=High, Wind=Strong? (1 mark)

(c) Convert the decision tree into a set of IF-THEN rules. (2 marks)

Question 52: Decision Tree Discussion (9 marks)

(a) Explain the information gain criterion used in ID3. Why does it prefer attributes with many values? (3 marks)

(b) Why does ID3 prefer shorter trees? How does this relate to generalization? (2 marks)

(c) Discuss the limitations of ID3 as a decision tree learning algorithm. (2 marks)

(d) What is the computational complexity of ID3? (2 marks)

Short Questions

Question 53: Entropy and Information Gain (9 marks)

(a) Define entropy. (2 marks)

(b) What is information gain and how is it calculated? (2 marks)

(c) Why is information gain used for attribute selection in ID3? (2 marks)

(d) Calculate the entropy of a set with 5 positive and 5 negative examples. (3 marks)

Question 54: ID3 Algorithm (9 marks)

(a) What is the basic idea of the ID3 algorithm? (2 marks)

(b) At each node, how does ID3 choose the attribute to split on? (2 marks)

(c) When does the ID3 algorithm stop growing the tree? (2 marks)

(d) Is the decision tree produced by ID3 unique for a given dataset? Explain. (3 marks)